

Linux Security & Scheduling – Practical Assignments

Topics Covered: Permissions, ACL, Special Permissions, chattr, and Scheduling

Assignment 1: Secure Department Shared Directory

1. Create the following groups: developers, testers, and managers.
2. Create users dev01, dev02, test01, test02, and manager01 and assign them to appropriate groups.

```
root@localhost:~# groupadd developers
root@localhost:~# groupadd testers
root@localhost:~# groupadd managers
root@localhost:~# getent group developers
developers:x:6671:
root@localhost:~# getent group testers
testers:x:6672:
root@localhost:~# getent group managers
managers:x:6673:
root@localhost:~# useradd -m -G developers dev01
root@localhost:~# useradd -m -G developers dev02
root@localhost:~# useradd -m -G testers test01
root@localhost:~# useradd -m -G testers test02
root@localhost:~# useradd -m -G managers manager01
root@localhost:~#
```

3. Create the directory /company/projects.
4. Configure permissions so developers have read, write, and execute access; testers have read and execute access; and managers have full access.

```
root@localhost:~# mkdir -p /company/projects
root@localhost:~# mkdir -p /backup/projects
root@localhost:~# chown root:developers /company/projects
root@localhost:~# chmod 2770 /company/projects
```

```
root@localhost:~# setfacl -m g:developers:rwx /company/projects
root@localhost:~# setfacl -m g:testers:rx /company/projects
root@localhost:~# setfacl -m g:managers:rwx /company/projects
root@localhost:~# setfacl -m o:--- /company/projects
```

5. Use ACLs to enforce the required access for the users/groups.
6. Configure a default ACL so that newly created files and directories inherit the required permissions.

```

root@localhost:~# setfacl -d -m g:testers:r-x /company/projects
root@localhost:~# setfacl -d -m g:developers:rwx /company/projects
root@localhost:~# setfacl -d -m g:managers:rwx /company/projects
root@localhost:~# setfacl -d -m u::rwx,g::rwx,o::--- /company/projects
root@localhost:~#

root@localhost:~# getfacl /company/projects
getfacl: Removing leading '/' from absolute path names
# file: company/projects
# owner: root
# group: developers
# flags: -s-
user::rwx
group::rwx
group:developers:rwx
group:testers:r-x
group:managers:rwx
mask::rwx
other::---
default:user::rwx
default:group::rwx
default:group:developers:rwx
default:group:testers:r-x
default:group:managers:rwx
default:mask::rwx
default:other::---

```

7. Apply SGID to /company/projects so that newly created files inherit the directory's group ownership.

```

root@localhost:~# chmod g+s /company/projects
root@localhost:~# ls -ld /company/projects
drwxrws---+ 2 root developers 6 Aug 24 11:45 /company/projects
root@localhost:~#

```

8. Apply the sticky bit where appropriate to prevent users from deleting files belonging to other users.

Command -- `chmod +t /company/projects`

9. Create the file /company/projects/project-plan.txt.

10. Use `chattr` to protect project-plan.txt from accidental modification or deletion.

11. Verify the file attributes using `lsattr` and verify permissions using `ls -ld` and `getfacl`.

```

bash: cd: /company: No such file or directory
root@localhost:~# cat /company/projects/project-plan.txt
Project Plan
root@localhost:~# chmod +i /company/projects/project-plan.txt
root@localhost:~# lsattr /company/projects/project-plan.txt
----i----- /company/projects/project-plan.txt
root@localhost:~# chmod -i /company/projects/project-plan.txt
root@localhost:~# lsattr /company/projects/project-plan.txt
----- /company/projects/project-plan.txt
root@localhost:~# ls -ld /company/projects
drwxrws--T+ 2 root developers 30 Aug 24 13:53 /company/projects
root@localhost:~# ls -l /company/projects/project-plan.txt
-rw-rw----+ 1 root developers 13 Aug 24 13:54 /company/projects/project-plan.txt
root@localhost:~# _

```

12. Create a scheduled job that performs a daily backup of /company/projects to /backup/projects/.

Command -- vi /usr/local/bin/project_backup.sh

Data under vi file:-

```
#!/bin/bash
```

```
DATE=$(date +%Y-%m-%d_%H-%M-%S)
```

```
mkdir -p /backup/projects
```

```
tar -czf /backup/projects/projects_$(DATE).tar.gz /company/projects
```

```

root@localhost:~# tar -czf backup.tar.gz /company/projects
tar: Removing leading '/' from member names
root@localhost:~# tar -tzf backup.tar.gz
company/projects/
company/projects/project-plan.txt

```

Test Backup Manually :

```

root@localhost:~# cd /usr/local/bin/
root@localhost:~# chmod +x /usr/local/bin/project_backup.sh
root@localhost:~# /usr/local/bin/project_backup.sh
tar: Removing leading '/' from member names
root@localhost:~# ls -lh /backup/projects/
total 4.0K
-rw-r--r-- 1 root root 195 Aug 24 15:44 projects_.tar.gz
root@localhost:~#

```

```

7:emp@CRONtab:~$ cd /usr/bin
crontab: installing new crontab
root@localhost:~# crontab -l
0 2 * * * /usr/local/bin/project_backup.sh
root@localhost:~#

```

```
root@localhost:~# ./project_backup.sh
tar: Removing leading `/' from member names
root@localhost:~# ls -lh /backup/projects/
total 4.0K
-rw-r--r-- 1 root root 195 Aug 24 15:52 projects_.tar.gz
root@localhost:~# tar -tzf /backup/projects/projects_.tar.gz
company/projects/
company/projects/project-plan.txt
root@localhost:~#
```

14. Create the users `appadmin`, `appuser`, `backupuser`, and `developer`.

```

root@localhost:~# which setfacl
/usr/bin/setfacl
root@localhost:~# which getfacl
/usr/bin/getfacl
root@localhost:~# useradd appadmin
root@localhost:~# useradd appuser
root@localhost:~# useradd backupuser
root@localhost:~# useradd developer
root@localhost:~# passwd appadmin
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
root@localhost:~# passwd appuser
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
root@localhost:~# passwd backupuser
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
root@localhost:~# passwd developer
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
root@localhost:~# id appadmin
uid=6675(appadmin) gid=6679(appadmin) groups=6679(appadmin)
root@localhost:~# id appuser
uid=6676(appuser) gid=6680(appuser) groups=6680(appuser)
root@localhost:~# id backupuser
uid=6677(backupuser) gid=6681(backupuser) groups=6681(backupuser)
root@localhost:~# id developer
uid=6678(developer) gid=6682(developer) groups=6682(developer)
root@localhost:~# _

```

15. Create the groups appadmins, appusers, and backupadmins and assign users appropriately.

```
root@localhost:~# groupadd appadmins
root@localhost:~# groupadd appusers
root@localhost:~# groupadd backupadmins
root@localhost:~# usermod -aG appadmins appadmin
root@localhost:~# usermod -aG appusers appuser
root@localhost:~# usermod -aG backupadmins backupuser
root@localhost:~# usermod -aG appusers developer
root@localhost:~# id appadmin
uid=6675(appadmin) gid=6679(appadmin) groups=6679(appadmin),6683(appadmins)
root@localhost:~# id appuser
uid=6676(appuser) gid=6680(appuser) groups=6680(appuser),6684(appusers)
root@localhost:~# id backupuser
uid=6677(backupuser) gid=6681(backupuser) groups=6681(backupuser),6685(backupadmins)
root@localhost:~# id developer
uid=6678(developer) gid=6682(developer) groups=6682(developer),6684(appusers)
root@localhost:~# getent group appadmins
appadmins:x:6683:appadmin
root@localhost:~# getent group appusers
appusers:x:6684:appuser,developer
root@localhost:~# getent group backupadmins
backupadmins:x:6685:backupuser
root@localhost:~#
```

16. Create the application directory /opt/myapp with the subdirectories config, logs, data, and scripts.

```
root@localhost:~# mkdir -p /opt/myapp/{config,logs,data,scripts}
root@localhost:~# ls -ld /opt/myapp
drwxr-xr-x. 6 root root 59 Aug 24 18:28 /opt/myapp
root@localhost:~# ls -ld /opt/myapp/*
drwxr-xr-x. 2 root root 6 Aug 24 18:28 /opt/myapp/config
drwxr-xr-x. 2 root root 6 Aug 24 18:28 /opt/myapp/data
drwxr-xr-x. 2 root root 6 Aug 24 18:28 /opt/myapp/logs
drwxr-xr-x. 2 root root 6 Aug 24 18:28 /opt/myapp/scripts
root@localhost:~#
```

17. Configure access so appadmin has full access; appuser has appropriate application access; developer has read access where required; and backupuser has backup-related read access.

```
root@localhost:~# setfacl -R -m u:appadmin:rwX /opt/myapp
root@localhost:~# setfacl -R -m u:appuser:rwX /opt/myapp
root@localhost:~# setfacl -R -m u:developer:rX /opt/myapp
root@localhost:~# setfacl -R -m u:backupuser:rX /opt/myapp
root@localhost:~# getfacl /opt/myapp
getfacl: Removing leading '/' from absolute path names
# file: opt/myapp
# owner: root
# group: root
user::rwX
user:appadmin:rwX
user:appuser:rwX
user:backupuser:r-X
user:developer:r-X
group::r-X
mask::rwX
other::r-X
root@localhost:~#
```

18. Use ACLs to implement the access requirements rather than relying only on traditional owner/group/other permissions.

19. Configure default ACLs where appropriate.

```

root@localhost:~# getfacl /opt/myapp
getfacl: Removing leading '/' from absolute path names
# file: opt/myapp
# owner: root
# group: root
user::rwx
user:appadmin:rwx
user:appuser:rwx
user:backupuser:r-x
user:developer:r-x
group::r-x
mask::rwx
other::r-x

root@localhost:~# setfacl -d -m u:appadmin:rwx,u:appuser:rwx,u:developer:r-x,u:backupuser:r-x /opt/myapp/config
root@localhost:~# getfacl /opt/myapp/config
getfacl: Removing leading '/' from absolute path names
# file: opt/myapp/config
# owner: root
# group: root
user::rwx
user:appadmin:rwx
user:appuser:rwx
user:backupuser:r-x
user:developer:r-x
group::r-x
mask::rwx
other::r-x
default:user::rwx
default:user:appadmin:rwx
default:user:appuser:rwx
default:user:backupuser:r-x
default:user:developer:r-x
default:group::r-x
default:mask::rwx
default:other::r-x

root@localhost:~#

```

20. Determine where SUID, SGID, and sticky bit should or should not be used and implement the appropriate special permissions.

```

root@localhost:~# chmod g+s /opt/myapp
root@localhost:~# ls -ld /opt/myapp
drwxrwsr-x+ 6 root root 59 Aug 24 18:28 /opt/myapp
root@localhost:~# find /opt/myapp -perm -4000
root@localhost:~# ls -ld /opt/myapp
drwxrwsr-x+ 6 root root 59 Aug 24 18:28 /opt/myapp
root@localhost:~#

```

21. Create /opt/myapp/config/application.conf and protect it using an appropriate chattr attribute.

```

root@localhost:~# touch /opt/myapp/config/application.conf
root@localhost:~# echo "applicatio.name=myapp" > /opt/myapp/config/application.conf
root@localhost:~# echo "applicatio.environment=production" >> /opt/myapp/config/application.conf
root@localhost:~# cat /opt/myapp/config/application.conf
applicatio.name=myapp
applicatio.environment=production
root@localhost:~# chattr +i /opt/myapp/config/application.conf
root@localhost:~# lsattr /opt/myapp/config/application.conf
---i----- /opt/myapp/config/application.conf
root@localhost:~#

```

22. Verify the file attribute using lsattr and test modification, deletion, and rename operations.

```

root@localhost:~# chattr +i /opt/myapp/config/application.conf
root@localhost:~# lsattr /opt/myapp/config/application.conf
-----i----- /opt/myapp/config/application.conf
root@localhost:~# echo "test=value" >> /opt/myapp/config/application.conf
-bash: /opt/myapp/config/application.conf: Operation not permitted
root@localhost:~# cat /opt/myapp/config/application.conf
applicatio.name=myapp
applicatio.environment=production
root@localhost:~# rm /opt/myapp/config/application.conf
rm: remove regular file '/opt/myapp/config/application.conf'? y
rm: cannot remove '/opt/myapp/config/application.conf': Operation not permitted
root@localhost:~# ls -l /opt/myapp/config/application.conf
-rw-rw-r--+ 1 root root 56 Aug 24 19:06 /opt/myapp/config/application.conf
root@localhost:~# mv /opt/myapp/config/application.conf /opt/myapp/config/application_backup.conf
mv: cannot move '/opt/myapp/config/application.conf' to '/opt/myapp/config/application_backup.conf': Operation not permitted
root@localhost:~# ls -l /opt/myapp/config/
total 4
-rw-rw-r--+ 1 root root 56 Aug 24 19:06 application.conf
root@localhost:~# chattr -i /opt/myapp/config/application.conf
root@localhost:~# lsattr /opt/myapp/config/application.conf
-----i----- /opt/myapp/config/application.conf
root@localhost:~# echo "applicatio.environment=testing" >> /opt/myapp/config/application.conf
root@localhost:~# cat /opt/myapp/config/application.conf
applicatio.name=myapp
applicatio.environment=production
applicatio.environment=testing
root@localhost:~#

```

```

root@localhost:~# id appadmin
uid=6675(appadmin) gid=6679(appadmin) groups=6679(appadmin),6683(appadmins)
root@localhost:~# id appuser
uid=6676(appuser) gid=6680(appuser) groups=6680(appuser),6684(appusers)
root@localhost:~# id backupuser
uid=6677(backupuser) gid=6681(backupuser) groups=6681(backupuser),6685(backupadmins)
root@localhost:~# id developer
uid=6678(developer) gid=6682(developer) groups=6682(developer),6684(appusers)
root@localhost:~# getent group appadmins
appadmins:x:6683:appadmin
root@localhost:~# getent group appusers
appusers:x:6684:appuser,developer
root@localhost:~# getent group backupadmins
backupadmins:x:6685:backupuser

```

```

root@localhost:~# ls -ld /opt/myapp
drwxrwsr-x+ 6 root root 59 Aug 24 18:28 /opt/myapp
root@localhost:~# ls -ld /opt/myapp/*
drwxrwxr-x+ 2 root root 30 Aug 24 18:57 /opt/myapp/config
drwxrwxr-x+ 2 root root 6 Aug 24 18:28 /opt/myapp/data
drwxrwxr-x+ 2 root root 6 Aug 24 18:28 /opt/myapp/logs
drwxrwxr-x+ 2 root root 6 Aug 24 18:28 /opt/myapp/scripts
root@localhost:~# getfacl /opt/myapp
getfacl: Removing leading '/' from absolute path names
# file: opt/myapp
# owner: root
# group: root
# flags: -s-
user::rwx
user:appadmin:rwx
user:appuser:rwx
user:backupuser:r-x
user:developer:r-x
group::r-x
mask::rwx
other::r-x
root@localhost:~# _

```

```

root@localhost:~# getfacl /opt/myapp/config
getfacl: Removing leading '/' from absolute path names
# file: opt/myapp/config
# owner: root
# group: root
user::rwx
user:appadmin:rwx
user:appuser:rwx
user:backupuser:r-x
user:developer:r-x
group::r-x
mask::rwx
other::r-x
default:user::rwx
default:user:appadmin:rwx
default:user:appuser:rwx
default:user:backupuser:r-x
default:user:developer:r-x
default:group::r-x
default:mask::rwx
default:other::r-x

```

```

root@localhost:~# ls -l /opt/myapp/config/application.conf
-rw-rw-r--+ 1 root root 87 Aug 24 19:25 /opt/myapp/config/application.conf
root@localhost:~# lsattr /opt/myapp/config/application.conf
----- /opt/myapp/config/application.conf
root@localhost:~#

```

23. Create an automated backup that runs every day at 11:00 PM and compresses /opt/myapp/config.

```

root@localhost:~# mkdir -p /backup/myapp
root@localhost:~# ls -ld /backup
drwxr-xr-x. 4 root root 59 Aug 24 19:44 /backup
root@localhost:~# ls -ld /backup/myapp
drwxr-xr-x. 2 root root 6 Aug 24 19:44 /backup/myapp
root@localhost:~# _

```

24. Store the backup under /backup/myapp/ using the filename format config-YYYY-MM-DD.tar.gz.

Command for Create the backup script -- nano /usr/local/bin/myapp_backup.sh

```

#!/bin/bash

BACKUP_DIR="/backup/myapp"
SOURCE_DIR="/opt/myapp/config"
DATE=$(date +%Y-%m-%d)

tar -czf "$BACKUP_DIR/config-$DATE.tar.gz" "$SOURCE_DIR"

find "$BACKUP_DIR" -type f -name "config-*.tar.gz" -mtime +7 -delete

```

Filename format -- cat /usr/local/bin/myapp_backup.sh

Make the script executable --

chmod +x /usr/local/bin/myapp_backup.sh

```

root@localhost:~# cat /usr/local/bin/myapp_backup.sh
#!/bin/bash

BACKUP_DIR="/backup/myapp"
SOURCE_DIR="/opt/myapp/config"
DATE=$(date +%Y-%m-%d)

tar -czf "$BACKUP_DIR/config-$DATE.tar.gz" "$SOURCE_DIR"

find "$BACKUP_DIR" -type f -name "config-*.tar.gz" -mtime +7 -delete
root@localhost:~# chmod +x /usr/local/bin/myapp_backup.sh
root@localhost:~# ls -l /usr/local/bin/myapp_backup.sh
-rwxr-xr-x. 1 root root 222 Aug 24 19:54 /usr/local/bin/myapp_backup.sh
root@localhost:~#

```

Test the backup manually first --

Do this before configuring cron.

```

root@localhost:~# /usr/local/bin/myapp_backup.sh
tar: Removing leading '/' from member names
root@localhost:~# ls -lh /backup/myapp/
total 4.0K
-rw-r--r--. 1 root root 212 Aug 24 20:05 config-2026-08-24.tar.gz
root@localhost:~#

```

25. Configure the backup process to delete backups older than 7 days.

Command : find "\$BACKUP_DIR" -type f -name "config-*.tar.gz" -mtime +7 -delete

26. Verify the permissions, ACLs, file attributes, and scheduled backup.

```

"/tmp/crontab.qX2Qzh" 2L, 44B written
crontab: installing new crontab
/root/.cache/crontab: mkdir: No such file or directory
root@localhost:~# ls -ld /opt/myapp
drwxrwsr-x+ 6 root root 59 Aug 24 18:28 /opt/myapp
root@localhost:~# ls -ld /opt/myapp/config
drwxrwsr-x+ 2 root root 30 Aug 24 18:57 /opt/myapp/config
root@localhost:~# ls -l /usr/local/bin/myapp_backup.sh
-rwxr-xr-x. 1 root root 222 Aug 24 19:54 /usr/local/bin/myapp_backup.sh
root@localhost:~# getfacl /opt/myapp/config
getfacl: Removing leading '/' from absolute path names
# file: opt/myapp/config
# owner: root
# group: root
user::rwx
user:appadmin:rwx
user:appuser:rwx
user:backupuser:r-x
user:developer:r-x
group::r-x
mask::rwx
other::r-x
default:user::rwx
default:user:appadmin:rwx
default:user:appuser:rwx
default:user:backupuser:r-x
default:user:developer:r-x
default:group::r-x
default:mask::rwx
default:other::r-x
root@localhost:~#

```

Verify ACLs :

For the configuration directory:

Command -- getfacl /opt/myapp/config

For the complete application directory:

Command -- getfacl -R /opt/myapp

```
# file: opt/myapp/config/application.conf
# owner: root
# group: root
user::rw-
user:appadmin:rw-          #effective:rw-
user:appuser:rw-          #effective:rw-
user:backupuser:r-x        #effective:r--
user:developer:r-x        #effective:r--
group:r-x                 #effective:r--
mask::rw-
other::r--

# file: opt/myapp/logs
# owner: root
# group: root
user::rwx
user:appadmin:rw-
user:appuser:rw-
user:backupuser:r-x
user:developer:r-x
group:r-x
mask::rwx
other:r-x

# file: opt/myapp/data
# owner: root
# group: root
user::rwx
user:appadmin:rw-
user:appuser:rw-
user:backupuser:r-x
user:developer:r-x
group:r-x
mask::rwx
other:r-x

# file: opt/myapp/scripts
# owner: root
# group: root
user::rwx
user:appadmin:rw-
user:appuser:rw-
user:backupuser:r-x
user:developer:r-x
group:r-x
mask::rwx
other:r-x

root@localhost:~#
```

For the backup directory:

Command -- getfacl /backup/myapp

```
root@localhost:~# getfacl /backup/myapp
getfacl: Removing leading '/' from absolute path names
# file: backup/myapp
# owner: root
# group: root
user::rwx
group::r-x
other::r-x

root@localhost:~# _
```

Verify file attributes :

For your protected configuration file:

Command -- lsattr /opt/myapp/config/application.conf

Verify the actual backup

After manually running the script:

Command -- ls -lh /backup/myapp/

inspect the archive without extracting it:

Command -- tar -tzf /backup/myapp/config-\$(date +%Y-%m-%d).tar.gz

```
root@localhost:~# lsattr /opt/myapp/config/application.conf
----- /opt/myapp/config/application.conf
root@localhost:~# ls -lh /backup/myapp/
total 4.0K
-rw-r--r--. 1 root root 212 Aug 24 20:05 config-2026-08-24.tar.gz
root@localhost:~# tar -tzf /backup/myapp/config-$(date +%y-%m-%d).tar.gz
tar (child): /backup/myapp/config-26-08-24.tar.gz: Cannot open: No such file or directory
tar (child): Error is not recoverable: exiting now
tar: Child returned status 2
tar: Error is not recoverable: exiting now
root@localhost:~#
```

Assignment 3: Linux Server Security & Compliance Automation

27. Create the users security01, auditor01, operator01, and admin01.

```

root@localhost:~# useradd security01
root@localhost:~# useradd auditor01
root@localhost:~# useradd operator01
root@localhost:~# useradd admin01
root@localhost:~# passwd security01
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
root@localhost:~# passwd auditor01
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
root@localhost:~# passwd operator01
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
root@localhost:~# passwd admin01
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
root@localhost:~# id security01
uid=6679(security01) gid=6686(security01) groups=6686(security01)
root@localhost:~# id auditor01
uid=6680(auditor01) gid=6687(auditor01) groups=6687(auditor01)
root@localhost:~# id operator01
uid=6681(operator01) gid=6688(operator01) groups=6688(operator01)
root@localhost:~# id admin01
uid=6682(admin01) gid=6689(admin01) groups=6689(admin01)
root@localhost:~#

```

28. Create the groups security, auditors, and operators and assign users appropriately.

```

root@localhost:~# groupadd security
root@localhost:~# groupadd auditors
root@localhost:~# groupadd operators

root@localhost:~# usermod -aG security security01
root@localhost:~# usermod -aG auditors auditor01
root@localhost:~# usermod -aG operators operator01
root@localhost:~# usermod -aG security admin01

```

```
root@localhost:~# id admin01
uid=6682(admin01) gid=6689(admin01) groups=6689(admin01),6690(security)
root@localhost:~# id operator01
uid=6681(operator01) gid=6688(operator01) groups=6688(operator01),6692(operators)
root@localhost:~# id auditor01
uid=6680(auditor01) gid=6687(auditor01) groups=6687(auditor01),6691(auditors)
root@localhost:~# id security01
uid=6679(security01) gid=6686(security01) groups=6686(security01),6690(security)
root@localhost:~#
```

29. Create the directory structure /security/confidential, /security/reports, /security/audit, and /security/scripts.

```
root@localhost:~# mkdir -p /security/{confidential,reports,audit,scripts}
root@localhost:~# ls -ld /security
drwxr-xr-x. 6 root root 69 Aug 25 09:49 /security
root@localhost:~# ls -ld /security/*
drwxr-xr-x. 2 root root 6 Aug 25 09:49 /security/audit
drwxr-xr-x. 2 root root 6 Aug 25 09:49 /security/confidential
drwxr-xr-x. 2 root root 6 Aug 25 09:49 /security/reports
drwxr-xr-x. 2 root root 6 Aug 25 09:49 /security/scripts
root@localhost:~#
```

30. Configure /security/confidential so admin01 and security01 have full access, auditor01 has read-only access, and operator01 has no access.

31. Implement the confidential directory access policy using ACLs and verify it using getfacl.

```

root@localhost:~# chown security01:security /security/confidential
root@localhost:~# chmod 770 /security/confidential
root@localhost:~# setfacl -m u:admin01:rwx /security/confidential
root@localhost:~# setfacl -m u:security01:rwx /security/confidential
root@localhost:~# setfacl -m u:auditor01:rx /security/confidential
root@localhost:~# setfacl -m u:operator01:--- /security/confidential
root@localhost:~# setfacl -d -m u:admin01:rwx /security/confidential
root@localhost:~# setfacl -d -m u:security01:rwx /security/confidential
root@localhost:~# setfacl -d -m u:auditor01:rx /security/confidential
root@localhost:~# setfacl -d -m u:operator01:--- /security/confidential
root@localhost:~# getfacl /security/confidential
getfacl: Removing leading '/' from absolute path names
# file: security/confidential
# owner: security01
# group: security
user::rwx
user:security01:rwx
user:auditor01:r-x
user:operator01:---
user:admin01:rwx
group::rwx
mask::rwx
other::---
default:user::rwx
default:user:security01:rwx
default:user:auditor01:r-x
default:user:operator01:---
default:user:admin01:rwx
default:group::rwx
default:mask::rwx
default:other::---

```

32. Configure /security/audit so multiple users can create files, users cannot delete files belonging to other users, and newly created files inherit the appropriate group.

33. Use the appropriate combination of SGID, sticky bit, and ACL and document the reason for each.

```

root@localhost:~# chown root:auditors /security/audit
root@localhost:~# chmod 3770 /security/audit
root@localhost:~# ls -ld /security/audit
drwxrws--T. 2 root auditors 6 Aug 25 09:49 /security/audit
root@localhost:~# stat -c '%A %a %U %G %n' /security/audit
drwxrws--T 3770 root auditors /security/audit

```

34. Create /security/confidential/security-policy.txt and protect it using chattr.

35. Verify the attribute using lsattr and test whether modification, deletion, and rename operations are permitted.

```
root@localhost:~# touch /security/confidential/security-policy.txt
root@localhost:~# chown security01:security /security/confidential/security-policy.txt
root@localhost:~# chmod 640 /security/confidential/security-policy.txt
root@localhost:~# chattr +i /security/confidential/security-policy.txt
root@localhost:~# lsattr /security/confidential/security-policy.txt
----i----- /security/confidential/security-policy.txt
root@localhost:~# echo "Security Policy" >> /security/confidential/security-policy.txt
-bash: /security/confidential/security-policy.txt: Operation not permitted
root@localhost:~# rm /security/confidential/security-policy.txt
rm: remove regular empty file '/security/confidential/security-policy.txt'? y
rm: cannot remove '/security/confidential/security-policy.txt': Operation not permitted
root@localhost:~# chattr -i /security/confidential/security-policy.txt
root@localhost:~# lsattr /security/confidential/security-policy.txt
----- /security/confidential/security-policy.txt
```

36. Create /usr/local/bin/security_check.sh to check permissions, ACL configuration, special permissions, chattr attributes, and ownership of important files.

Command – vi /usr/local/bin/security_check.sh (Create the script)

```

LOG="/var/log/security_check.log"

{
echo "=====
echo "Security Compliance Check"
echo "Date: $(date)"
echo "Hostname: $(hostname)"
echo "=====

echo
echo "Directory Permissions:"
ls -ld /security
ls -ld /security/confidential
ls -ld /security/reports
ls -ld /security/scripts

echo
echo "ACL Configuration:"
getfacl /security/confidential
getfacl /security/audit

echo
echo "Special Permissions:"
stat -c '%A %a %U %G %n' /security/audit

echo
echo "Protected Files:"
lsattr /security/confidential/security-policy.txt

echo
echo "Important File Ownership:"
ls -l /security/confidential/security-policy.txt

echo
echo "Security Check Completed"
echo "=====
} >> "$LOG" 2>&1

~
root@localhost:~# chmod 750 /usr/local/bin/security_check.sh
root@localhost:~# ls -l /usr/local/bin/security_check.sh
-rwxr-x---. 1 root root 809 Aug 25 11:13 /usr/local/bin/security_check.sh

```

37. Configure the script to write its results to /var/log/security_check.log.

Commands – touch /var/log/security_check.log
 chown root:root /var/log/security_check.log
 chmod 640 /var/log/security_check.log
 /usr/local/bin/security_check.sh
 cat /var/log/security_check.log

```

Directory Permissions:
drwxr-xr-x. 6 root root 69 Aug 25 09:49 /security
drwxrwx---+ 2 security01 security 33 Aug 25 10:15 /security/confidential
drwxr-xr-x. 2 root root 6 Aug 25 09:49 /security/reports
drwxr-xr-x. 2 root root 6 Aug 25 09:49 /security/scripts

ACL Configuration:
getfacl: Removing leading '/' from absolute path names
# file: security/confidential
# owner: security01
# group: security
user::rwx
user:security01:rwx
user:auditor01:r-x
user:operator01:---
user:admin01:rwx
group::rwx
mask::rwx
other::---
default:user::rwx
default:user:security01:rwx
default:user:auditor01:r-x
default:user:operator01:---
default:user:admin01:rwx
default:group::rwx
default:mask::rwx
default:other::---

getfacl: Removing leading '/' from absolute path names
# file: security/audit
# owner: root
# group: auditors
# flags: -st
user::rwx
group::rwx
other::---

Special Permissions:
drwxrws--T 3770 root auditors /security/audit

Protected Files:
----- /security/confidential/security-policy.txt

Important File Ownership:
-rw-r-----+ 1 security01 security 0 Aug 25 10:15 /security/confidential/security-policy.txt

Security Check Completed

```

38. Schedule the security compliance check to run every day at 01:00 AM.

```

root@localhost:~# crontab -l

0 1 * * * /usr/local/bin/security_check.sh
root@localhost:~#

```

39. Ensure the log records the date, hostname, directory permissions, ACL status, special permissions, and protected files.
40. Demonstrate the complete security configuration and scheduled compliance check.

```

root@localhost:~# grep "Date:" /var/log/security_check.log
Date: Tue 25 Aug 13:01:50 IST 2026
Date: Tue 25 Aug 13:09:01 IST 2026
root@localhost:~# grep "Hostname:" /var/log/security_check.log
Hostname: localhost.localdomain
Hostname: localhost.localdomain

root@localhost:~# grep -A6 "Directory Permissions:" /var/log/security_check.log
Directory Permissions:
drwxr-xr-x. 6 root root 69 Aug 25 09:49 /security
drwxrwx---+ 2 security01 security 33 Aug 25 10:15 /security/confidential
drwxr-xr-x. 2 root root 6 Aug 25 09:49 /security/reports
drwxr-xr-x. 2 root root 6 Aug 25 09:49 /security/scripts

ACL Configuration:
--
Directory Permissions:
drwxr-xr-x. 6 root root 69 Aug 25 09:49 /security
drwxrwx---+ 2 security01 security 33 Aug 25 10:15 /security/confidential
drwxr-xr-x. 2 root root 6 Aug 25 09:49 /security/reports
drwxr-xr-x. 2 root root 6 Aug 25 09:49 /security/scripts

```

```
ACL Configuration:
root@localhost:~# grep -A20 "ACL Configuration:" /var/log/security_check.log
ACL Configuration:
getfacl: Removing leading '/' from absolute path names
# file: security/confidential
# owner: security01
# group: security
user::rwx
user:security01:rwx
user:auditor01:r-x
user:operator01:---
user:admin01:rwx
group::rwx
mask::rwx
other::---
default:user::rwx
default:user:security01:rwx
default:user:auditor01:r-x
default:user:operator01:---
default:user:admin01:rwx
default:group::rwx
default:mask::rwx
default:other::---
--
ACL Configuration:
getfacl: Removing leading '/' from absolute path names
# file: security/confidential
# owner: security01
# group: security
user::rwx
user:security01:rwx
user:auditor01:r-x
user:operator01:---
user:admin01:rwx
group::rwx
mask::rwx
other::---
default:user::rwx
default:user:security01:rwx
default:user:auditor01:r-x
default:user:operator01:---
default:user:admin01:rwx
default:group::rwx
default:mask::rwx
default:other::---
```

```
root@localhost:~# grep -A2 "Special Permissions:" /var/log/security_check.log
Special Permissions:
drwxrws--T 3770 root auditors /security/audit

--
Special Permissions:
drwxrws--T 3770 root auditors /security/audit

root@localhost:~# grep -A2 "Protected Files:" /var/log/security_check.log
Protected Files:
----- /security/confidential/security-policy.txt

--
Protected Files:
----- /security/confidential/security-policy.txt

root@localhost:~# _
```